

Myelin Water Fraction and Intra/Extracellular Water Geometric Mean T_2 Normative Atlases for the Cervical Spinal Cord from 3T MRI

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ABSTRACT

BACKGROUND AND PURPOSE: Acquiring and interpreting quantitative myelin-specific MRI data at an individual level is challenging because of technical difficulties and natural myelin variation in the population. To overcome these challenges, we used multiecho T_2 myelin water imaging (MWI) to create T_2 metric healthy population atlases that depict the mean and variation of myelin water fraction (MWF), and intra- and extracellular water mobility as described by geometric mean T_2 (IEGMT₂).

METHODS: Cervical cord MWI was performed at 3T on 20 healthy individuals (10M/10F, mean age: 36 years) and 3 relapsing remitting multiple sclerosis (RRMS) participants (1M/2F, age: 39/42/37 years). Anatomical data were collected for the purpose of image segmentation and registration. Atlases were created by coregistering and averaging T_2 metrics from all controls. Voxel-wise z-score maps from 3 RRMS participants were produced to demonstrate the preliminary utility of the MWF and IEGMT₂ atlases.

RESULTS: The average MWF atlas provides a representation of myelin in the spinal cord consistent with well-known spinal cord anatomical characteristics. The IEGMT₂ atlas also depicted structural variations in the spinal cord. Z-score analysis illustrated distinct abnormalities in MWF and IEGMT₂ in the 3 RRMS cases.

CONCLUSIONS: Our findings highlight the potential for using a quantitative T_2 relaxation metric atlas to visualize and detect pathology in spinal cord. Our MWF and IEGMT₂ atlases (URL: <https://sourceforge.net/projects/mwi-spinal-cord-atlases/>) can serve as normative references in the cervical spinal cord for other studies.

Keywords: atlas, cervical spinal cord, magnetic resonance imaging, myelin water imaging, white matter.

Acceptance: Received May 30, 2019, and in revised form July 19, 2019. Accepted for publication July 23, 2019.

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Acknowledgments and disclosure: We sincerely thank the study participants and MRI technologists at the UBC MRI Research Centre. We gratefully acknowledge funding support from an International Collaboration on Repair Discoveries (ICORD) Seed Grant and an MS Society of Canada Operating Grant. HL is the recipient of an end MS PhD Scholarship. ELM is supported by Philips Canada. JLKK holds operating grants from CIHR and the Natural Sciences and Engineering Research Council of Canada (NSERC). SHK and CL hold operating grants from the Multiple Sclerosis Society of Canada and NSERC.

HL, AD, LB, have nothing to disclose. EL is in receipt of a PhD studentship jointly funded by General Electric (GE) Healthcare and the National Institute for Health Research (NIHR) Biomedical Research Centre at South London and Maudsley NHS Foundation Trust and King's College London. The views expressed are those of the author and not necessarily those of the NHS, the NIHR or the Department of Health and Social Care. ELM receives salary support from Philips Canada. DKBL has received research funding from the Canadian Institute of Health Research and Multiple Sclerosis Society of Canada. He is the Emeritus Director of the UBC MS/MRI Research Group which has been contracted to perform central analysis of MRI scans for therapeutic trials with Novartis, Perceptives, Roche and Sanofi-Aventis. The UBC MS/MRI Research Group has also received grant support for investigator-initiated independent studies from Genzyme, Merck-Serono, Novartis and Roche. He has acted as a consultant to Vertex Pharmaceuticals and served on the Data and Safety Advisory Board for Opexa Therapeutics and Scientific Advisory Boards for Adelphi Group, Novartis and Roche. He has also given lectures which have been supported by nonrestricted education grants from Novartis and Biogen. AT has received research grants from the MS Society of Canada and honoraria from Bayer Pharmaceuticals, Teva Neurosciences, BioMS, Bayhill, and EMD Serono not related to this analysis. SHK receives research funding from Roche, Genzyme, the Vancouver Coastal Health Research Institute, the Milan & Maureen Ilich Foundation, the Multiple Sclerosis Society of Canada, the Natural Sciences and Engineering Research Council of Canada, and the Michael Smith Foundation for Health Research. JLKK receives research funding from the Natural Sciences and Engineering Research Council of Canada. CL receives research funding from the MS Society of Canada, the International Collaboration on Repair Discoveries, and the Natural Sciences and Engineering Research Council of Canada.

J Neuroimaging 2020;30:50-57.

DOI: 10.1111/jon.12659

Introduction

Myelin is an insulating sheath around central nervous system (CNS) axons, which is crucial for rapid signal conduction.¹ Myelin damage or loss can occur in spinal cord injuries (SCI)

or neurodegenerative diseases, leading to impaired function of the CNS. Quantitative in vivo measurement of myelin is thus highly desirable to better assess the prognosis, severity and progression of spinal cord injury and disease, and may provide

valuable guidance for treatment planning. However, reliably detecting the change of myelin content on an individual level is still very challenging for a number of reasons.

The first challenge comes from technical difficulties in detecting *in vivo* myelin in the spinal cord. Several advanced magnetic resonance imaging (MRI) techniques have been developed to quantify the myelin content in the CNS, such as diffusion tensor imaging (DTI) and magnetization transfer imaging (MTI).^{2–5} Applying these techniques in the spinal cord^{6–10} is intrinsically more difficult than in brain due to the possible artifacts from physiological motion and small cross-sectional dimensions of the cord, and inhomogeneous magnetic field in this region.¹¹ These factors make individual imaging results less reliable and more difficult to interpret. For this reason, group comparisons are often used in many studies to detect the differences between control and disease cohorts. In addition, while DTI and MTI are undoubtedly sensitive to pathological changes, they are affected not only by changes in myelin but also other factors including edema, inflammation, and gliosis.¹²

Another challenge is that normal myelin variability in the healthy population is not well characterized. Myelination has been reported to vary with age.¹ Other factors such as sex, laterality, and ethnicity may also have unknown effects on the formation and development of normal myelin. These possibly myelin-related factors make the interpretation of individual imaging results more difficult. In a clinical setting, it is not feasible to scan a perfectly matched healthy control for every individual patient. Furthermore, collecting control data for every clinical or research study is costly, diverting funds from acquiring more data on patients.

Unfortunately, the above challenges cannot be fully addressed on an individual level with current technology. To mitigate these challenges we propose to create an *in vivo* myelin atlas for the spinal cord. Since the atlas is essentially an average image of many healthy subjects, signal to noise ratio (SNR) of the atlas will be drastically improved, compared to a single individual, and the atlas is expected to show superior image quality and thus greater details about myelin distribution. In addition, atlas creation enables quantification of the variations (standard deviations) of myelin among a healthy population. Ultimately, the created atlas together with estimated variations in population can be used as a reference, for comparing individual patients.

For this work, we used multiecho T_2 myelin water imaging (MWI)^{13,14} to quantitatively and specifically examine the myelin content in the cervical spinal cord.¹⁵ The MR signal from myelin water, the water trapped between myelin bilayers, can be extracted from the total MR signal based on a characteristic short T_2 relaxation time. The ratio of myelin water signal relative to the total signal is termed myelin water fraction (MWF), which is used as a biomarker for myelin content. The myelin specificity of MWF has been histologically validated in human brain and spinal cord as well as in animal models.^{16–19} The data collection and analysis required for MWF also provides the geometric mean T_2 , which represents the geometric mean (amplitude-weighted mean on a logarithmic scale)²⁰ of the T_2 times of the intra- and extracellular (IE) water (IEGMT₂) and is often used as complementary information to the MWF results. It is believed that the IEGMT₂ is influenced by intra- and extra-axonal morphology and environment,^{21–23} and in pathol-

ogy, IEGMT₂ is typically reported to increase, likely due to the presence of extra water from inflammation or edema.^{24–28}

Besides multiecho T_2 based MWI, a number of other quantitative techniques use fundamentally different approaches to extract MWF and IEGMT₂ values. For example, the multi-component driven equilibrium single pulse observation of T_1 and T_2 (mcDESPOT) method makes use of T_1 weighted and steady state gradient echo acquisitions to estimate T_1 and T_2 times of the tissue components.²⁹ And the multiecho gradient echo (mGRE) technique collects T_2^* decay signals to carry out the MWF.³⁰ These approaches generally offer a larger coverage within shorter scanning time, but the data analysis is usually performed under the restricted assumption of a 2- or 3-pool model and requires estimation of a large number of parameters from a limited amount of data points. Consequently, the accuracy of MWF derived from these techniques (eg, mcDESPOT and mGRE) is presently an active area of research.^{31–33} Although multiecho T_2 based MWI has a few limitations such as smaller coverage, the imaging protocols and postprocessing analysis of this method have been well established over the years and are widely used in numerous studies of myelin quantification, making it an appropriate choice for our study.

The goal of our study was to create multiecho T_2 -derived MWF and IEGMT₂ atlases, by coregistering and averaging MWF and IEGMT₂ maps of the cervical spinal cord from healthy subjects. These 2 quantitative atlases, with estimated population variations, can serve as normative references of the cervical spinal cord for future studies. We demonstrated the utility of these atlases by applying them to detect pathology in the spinal cords of individual multiple sclerosis patients.

Methods

Subject Information

Twenty healthy controls (10 male/10 female; mean age 36 years, range 21–70 years) and 3 participants with relapsing remitting multiple sclerosis (RRMS subject 1: 39-year-old male, disease duration: 5 years, Expanded Disability Status Scale, EDSS: 2.5; RRMS subject 2: 42-year-old female, disease duration: 4.5 years, EDSS: 0; RRMS subject 3: 37-year-old female, disease duration: 7 years, EDSS: 4.0) were recruited. All subjects provided written informed consent with approval from the local ethics board.

MRI Experiments

Subjects were scanned on a 3.0T Achieva MRI system (Philips, Best, the Netherlands) with a 6 channel phased array spine coil. Cervical cord localizer and sagittal T_2 -weighted sequences were followed by (1) axial high-resolution multiecho fast field echo (mFFE, TE1/ Δ TE/TR = 6.6/8.2/815 ms, 5 echoes cumulated, field of view (FOV) = 150 mm $\times$ 150 mm $\times$ 44 mm, resolution = 0.8 mm $\times$ 0.8 mm $\times$ 2.5 mm) for anatomical white and gray matter contrast and (2) a 3D gradient and spin echo (GRASE) T_2 relaxation experiment (excitation flip angle = 90 degrees, 32-echo, TE1/ Δ TE/TR = 10/10/1500 ms, refocusing flip angle = 180 degrees, 8 axial slices, oversampling = 1.3, gradient mode = maximum, gradient performance = 40 mT/m at 200 mT/m/ms, fold-over direction = RL, EPI factor = 3, fat suppression = no, presaturation band = no, FOV = 180 mm $\times$ 150 mm $\times$ 40 mm, resolution = 0.75 mm $\times$ 0.75 mm $\times$ 5 mm).¹⁵ The slab was centered at the C2/C3 cord level for all subjects.

Myelin Water Fraction and Geometric Mean T_2

Voxel-wise decay curve analysis was performed using regularized nonnegative least squares (NNLS) with the extended phase graph algorithm to estimate the refocusing flip angle in each voxel and correcting the T_2 decay curve for stimulated echo artifacts.³⁴ The regularized NNLS analysis was initialized by using 40 T_2 exponentials logarithmically spaced from 15 ms to 2 s to fit the decay data. The misfit χ^2 is minimized by the following formula: $\sum_{i=1}^N |\sum_{j=1}^M A_{ij}s_j - y_i|^2 + \mu \sum_{i=1}^N |s_j|^2$, where N represents the total number of data points; M represents the number of logarithmically spaced T_2 decay times, s_i represents the amplitude of the T_2 distribution that needs to be solved; A_{ij} is the kernel matrix for the T_2 exponential decay; y_i is the measurement data; μ is the regularization factor and was chosen to be 1.02 in our analysis.^{14,35,36} MWF was defined as the fractional signal with T_2 less than 40 ms, which was chosen in accordance with previous cervical spinal cord studies using GRASE MWI.^{15,37} IEGMT₂ was the amplitude-weighted mean T_2 time of the intra- and extracellular water compartment and was calculated by the following formula: $\text{IEGMT}_2 = \frac{\sum_{i=1}^N S_i \times T_i}{\sum_{i=1}^N S_i}$, where N is the total number of T_2 components logarithmically spaced between 40 – 200 ms; T_i is the T_2 time of the i^{th} component; and S_i is the amplitude of the i^{th} component. Voxel-wise MWF and IEGMT₂ maps were used for subsequent analysis. The NNLS analysis code can be requested from: <https://mrresearch.med.ubc.ca/news-projects/myelin-water-fraction/>.

Myelin Water Fraction Atlas and Geometric Mean T_2 Atlases

For creation of T_2 relaxation metric atlases, individual MWF and IEGMT₂ maps were coregistered to a common template space. We used the Spinal Cord Toolbox (SCT)³⁸ for image postprocessing and the PAM50 template³⁹ as the common space for atlas creation. A registration framework based on each individual's white matter morphology was developed for atlas creation:

1. The anatomical mFFE image was registered to GRASE space by an affine transformation;
2. A gray matter mask was manually drawn on each slice of the registered mFFE image;
3. A white matter mask was calculated by subtracting the manual gray matter segmentation from the whole cord mask (obtained using the SCT propseg tool);
4. The white matter mask was registered to the PAM50 white matter template using a nonlinear transformation (a combination of center of mass rotation, affine, symmetric normalization, and bsplinesyn algorithms) to obtain the warping function which transforms individual GRASE images to the template space;
5. The acquired warping function was applied to individual MWF and IEGMT₂ maps to obtain the registered MWF and IEGMT₂ maps in template space. Registered images were visually inspected.

The atlases were then created by averaging the coregistered MWF and IEGMT₂ maps from all individual subjects in the template space. The voxel-wise standard deviation (SD) of each atlas was subsequently calculated to assess the variations of MWF and IEGMT₂ in this healthy population.

Z-Score Maps for RRMS Subjects

To illustrate the utility of the created atlases in disease, voxel-wise z -score maps of 3 RRMS subjects were calculated as

$z = \frac{x - \mu}{\sigma}$, where x is the individual measurements of MWF or IEGMT₂ maps registered in the template space, μ is MWF or IEGMT₂ atlas (the mean), and σ is the standard deviation (SD) of the atlas. Regions with coefficient of variation (COV: SD divided by mean) larger than 0.75 were excluded.⁴⁰ Z -score map images were visually inspected to mimic typical clinical review for abnormalities which illustrates how the atlases can be used to detect abnormalities in individual patients. In addition, significant differences were defined as $z < -1.96$ or $z > 1.96$.

Results

Myelin Water Fraction Atlas and Geometric Mean T_2 Atlases

Figure 1 shows the resulting MWF atlas and IEGMT₂ atlas as well as their SD maps of all 8 slices from C2 to C3. Visual inspection of the atlas maps demonstrates that distinct regions are easily identifiable, especially in the MWF atlas. More dramatic variations along the cord from C2 to C3 were observed in the MWF atlas compared to the IEGMT₂ atlas. The SD of the MWF atlas was homogenous (Fig 1) over the cross-sectional area of the whole cord, whilst the SD of IEGMT₂ atlas showed slight gray and white matter contrast.

Z-Score Maps for RRMS Participants

To demonstrate the utility of the atlases, anatomical mFFE, MWF and IEGMT₂ z -score maps from 3 RRMS participants are presented in sagittal, coronal and axial views (Fig 2). Noisy individual MWF and IEGMT₂ maps may introduce false positive points so thresholded maps ($z < -1.96$ or $z > 1.96$) are also displayed to show regions with significant differences compared to the atlases. In the MWF z -score maps (Fig 2A), 3 subjects showed distinct characteristics in MWF z -scores. Significant myelin water signal reductions ($z < -1.96$) were found in the lesions of RRMS subject 1 and 2. Particularly for RRMS subject 1, low MWF z -scores within the dorsal column regions across the entire C2 and C3 segments were also observed on the averaged axial slice. Subject 2 and 3 showed a lower degree of abnormalities in the axial averaged MWF z -score map. IEGMT₂ z -score maps (Fig 2B) showed no evidence of large changes in the averaged IEGMT₂ for any subject. Interestingly, IEGMT₂ was increased slightly in the lesion of RRMS subject 1 but not in the lesion of RRMS subject 2.

Discussion

Although MWI is a robust method for myelin content quantification, and a good scan-rescan reproducibility has previously been reported,¹⁵ MWF values of healthy individuals can vary due to many factors (eg, normal population variation, factors affecting data quality and complexity of postprocessing analysis), making the image quality of individual spinal cord MWF maps less than ideal. The idea of creating a MWF atlas takes advantage of the fact that averaging a large number of samples can minimize random noise and improve SNR. Indeed, the resulting MWF and IEGMT₂ atlases demonstrated much improved image quality compared to individual metric maps (Fig 3). However, we have to point out that MWF and IEGMT₂ values are also dependent on factors regarding experiment protocol and data analysis. Therefore, following the same experimental and postprocessing parameters aforementioned in the Methods is important when utilizing our atlases.

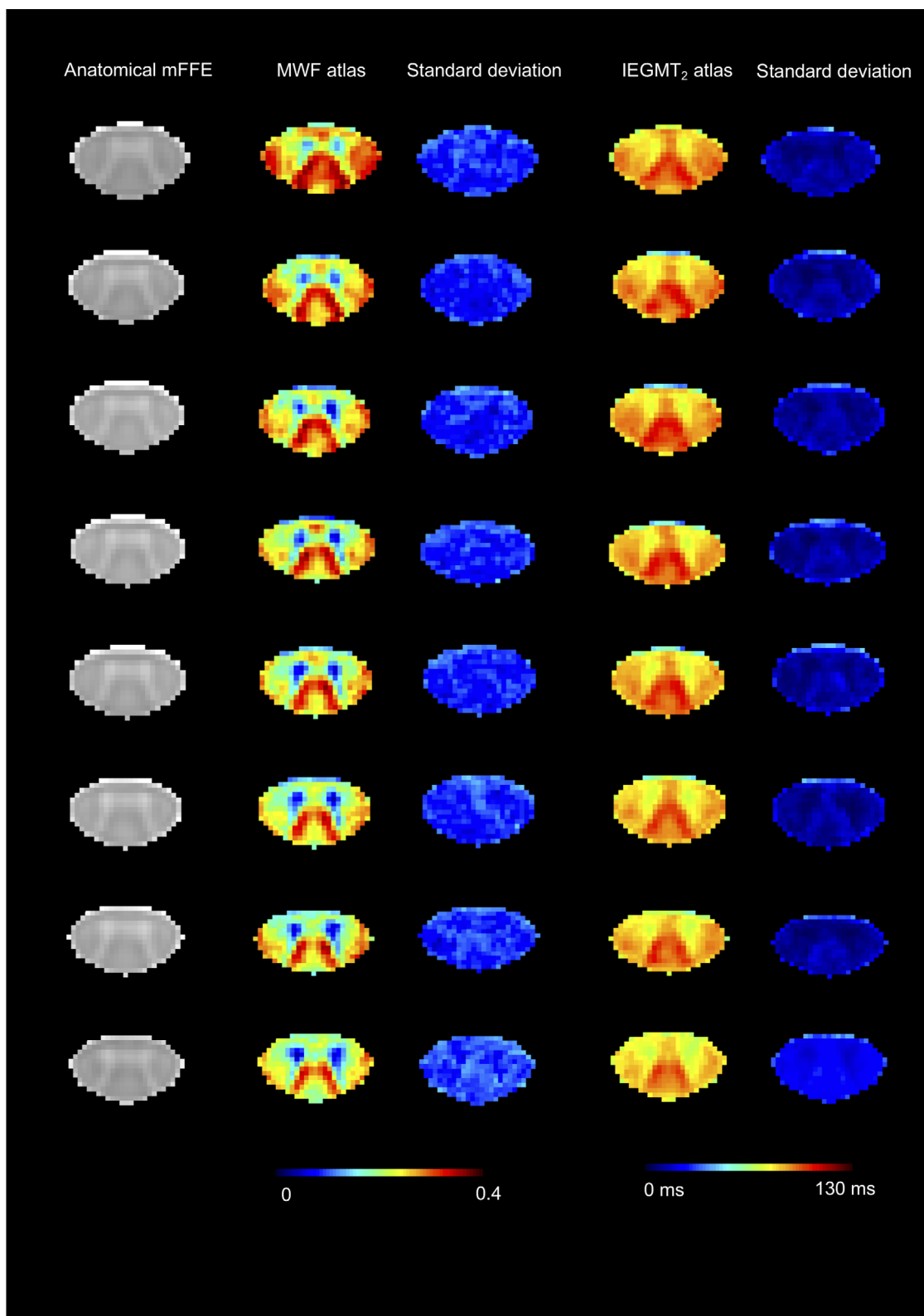


Fig 1. MWF atlas and IEGMT₂ atlas. The resulting myelin water fraction (MWF) atlas and geometric mean T₂ (IEGMT₂) atlas of all 8 slices from C2 to C3. Averaged anatomical images (mFFE) are also displayed.

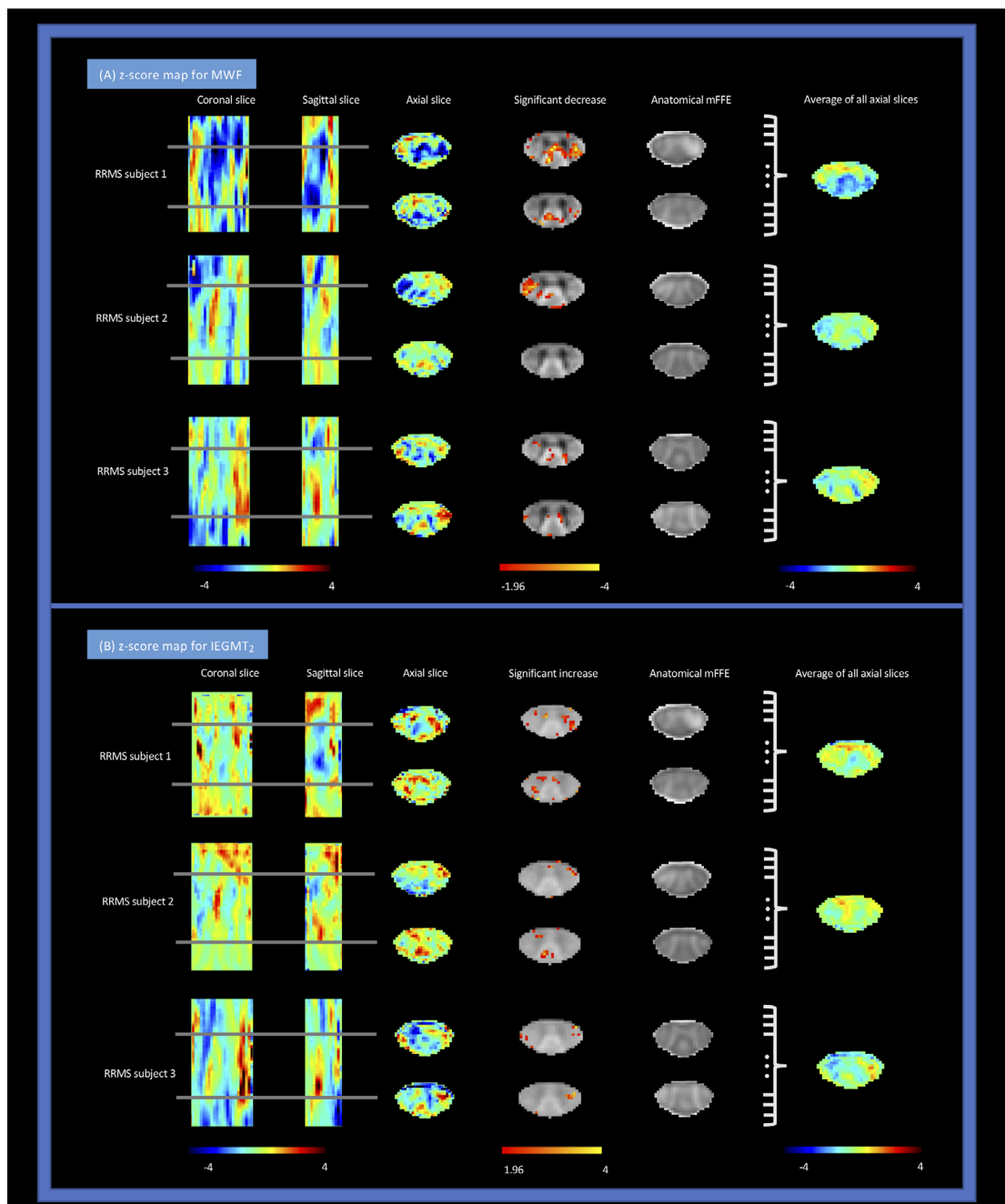


Fig 2. Comparison of z-score maps from 3 relapsing remitting multiple sclerosis (RRMS) subjects. Z-score maps were calculated based on myelin water fraction (MWF) and geometric mean T2 (IEGMT₂) atlases and their standard deviation for C2 to C3 cord and representative slices are presented in sagittal, coronal, and axial views. Sagittal and coronal views are showing at the middle of the cord, and axial view is showing at C2 and C3 levels. There are 50 registered axial slices in the rostral-caudal direction in template space. Axial slices are also thresholded for regions of significance ($z < -1.96$ or $z > 1.96$) and displayed on MWF and IEGMT₂ atlases respectively. An average of all 50 axial z-score maps is also shown for each participant. Axial anatomical multiecho fast field echo (mFFE) images are shown to compare with the z-score maps at different levels.

The MWF atlas accurately represents well-known myelin anatomical characteristics.^{41–43} Anatomical features of the cord such as the butterfly shaped gray matter, dorsal columns, and several lateral and ventral pathways are clearly depicted on the atlas (Fig 1), supporting the usefulness of the atlas as an in vivo myelin reference in the spinal cord. Recent studies^{44,45} using proton density imaging to estimate the macromolecular tissue volume (MTV) found similar variations between different spinal cord regions. Also from those studies, average maps

based on diffusion and MT metrics exhibited different regional variations in the cord, indicating that enriched tissue properties (such as axon diameter, myelin thickness, and water mobility, etc.) can be probed and monitored differently. It seems that using any single quantitative imaging method alone may only provide one aspect of tissue properties and a combination of different techniques may offer a better understanding of axonal morphology and myelin integrity. It is also interesting to observe that, within only 2 vertebral levels of the cervical cord

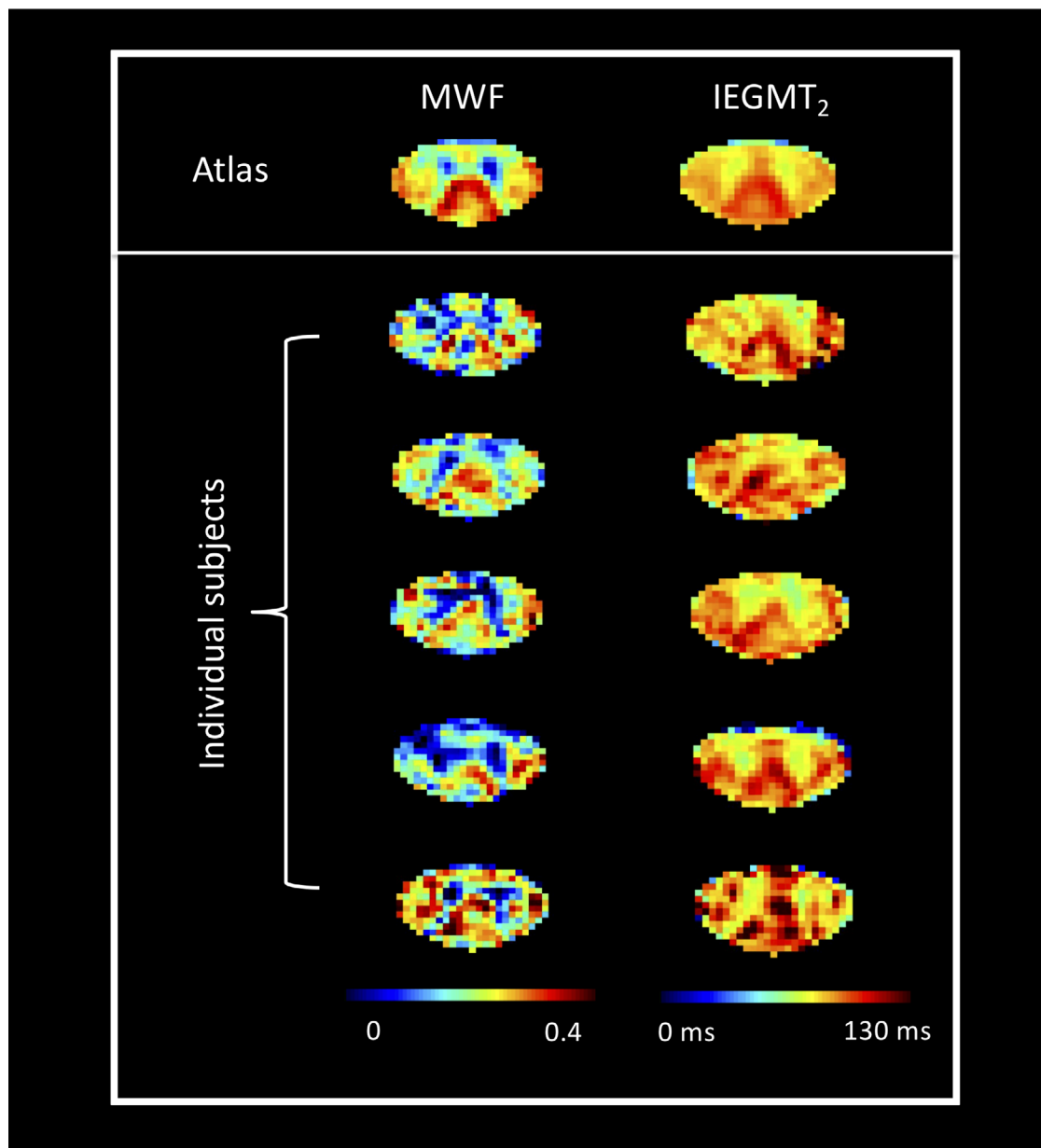


Fig 3. Comparison of MWF and IEGMT₂ atlases with individual metric maps. In this illustrative example, the created atlases showed much better image quality and contrast between structures compared to individual myelin water fraction (MWF) and geometric mean T₂ (IEGMT₂) maps of 5 subjects. All slices are shown at the same level.

that we imaged, the atlas shows distinct regional variation patterns (Fig 1) from C2 to C3, revealing how much the myelin may change through the cord even over a small distance. The changes of myelin along the cord is supported by Minty et al who, using a single slice sagittal acquisition at 1.5 T, observed variations from C2 to C3, as well as along the rest of the spinal cord.⁴⁶

In contrast, the IEGMT₂ atlas shows minimal variation along the cord from C2 to C3, indicating that IEGMT₂ offers additional and different information about the cord tissue morphology. Although the distinction between regions is not as prominent as in the MWF atlas, many spinal cord regions can still be differentiated on the IEGMT₂ atlas. Histological validation for axonal diameter and density will be needed for further interpretations of IEGMT₂ results.

Z-score map (Fig 2) of an individual subject provides a useful statistical parameter and is one of the potential applications of the MWF and IEGMT₂ atlases. Z-score maps can detect true abnormalities by accounting for normal tissue variations within healthy populations, and have been successful in brain imaging applications.⁴⁷ Our representative examples from 3 RRMS subjects show that the severity and location of demyelination and alterations in intra/extracellular water mobility in the cord can be clearly visualized in the resulting z-score maps. One may also notice that a few small highlighted regions in the z-score map do not correspond to lesions, which confounds the interpretation of the z-score maps. Such discrepancy is probably resulting from the high variations in individual MWF and IEGMT₂ values due to noise and NNLS instability. A potential solution is to incorporate nonlocal regularization filters^{48,49}

into the current analysis pipeline to denoise the GRASE images. However, these nonlocal regularization methods have only been validated for brain imaging data so feasibility in the spinal cord needs further investigation. In summary, the z -score map is a straightforward way to make use of T_2 metric atlases as a reference to visualize pathology in individual subjects.

A number of limitations warrant consideration. First, at the moment, only the cervical cord is accessible for good quality data acquisition using MWI. But for many diseases and injuries, spinal cord pathology can occur also, as well as only, in thoracic regions. However, more artifacts are generated due to technical difficulties from more cord motion near the lungs and heart, larger imaging volume, and more main field inhomogeneity, making investigations at the thoracic level very challenging. Particularly, the reduction of MWI accessibility below C7 is largely due to the fact that the imaging field of view has to be substantially increased to prevent any phase wrapping from the shoulders, leading to an impractical scanning time if the same resolution is preserved. The challenges are common issues for all quantitative MR techniques in the spinal cord. Second, although automatic registration and segmentation by SCT can exclude human bias, the quality of these image processing steps still needs further validation, given the fact that the spinal cord is so small that any misalignment in cord registration may alter the results of small regions within the cord. Thirdly, the atlas is generated from a limited number of healthy subjects. As age has been reported as a significant factor in spinal cord MWF measurement in a previous study,⁵⁰ it would be important to follow-up the present study by recruiting a much larger number of healthy subjects spanning all ages to enable creation of a more extensive atlas for specific age and sex groups. Finally, a few other conditions such as edema or inflammation could also lead to a decrease in MWF values and complicate the interpretation of MWF reduction. Although ex vivo imaging and histological validation¹⁷ suggests a linear relationship between MWF and myelin specific stains such as Luxol Fast Blue, the total water content is still a confounding factor for in vivo imaging. Previously, a model created to interpret experimental studies in MS brains⁵¹ found that the increase in water content needed to yield the substantial reductions in MWF observed would require a volume of water that would lead to physiologically unrealistic tissue swelling. Nevertheless, a measurement of the total water content^{52,53} in addition to MWI would certainly be helpful to discriminate myelin loss from other influencing conditions.

In conclusion, we have created an in vivo normative atlas of quantitative T_2 relaxation metrics for the spinal cord. The MWF atlas characterizes myelin content in distinct regions while the IEGMT₂ atlas provides information about intra and extracellular water mobility. Both atlases can be used as a reference for other in vivo spinal cord studies that assess demyelination and other pathologies like inflammation which may impact IEGMT₂ of individual patients (URL: <https://sourceforge.net/projects/mwi-spinal-cord-atlases/>), an important step toward the ultimate goal of clinical implementation and use of quantitative T_2 relaxation imaging.

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